LESSON 5-4
Solving Radical Equations

Lesson Overview

FOCUS

Objective

Students will be able to:

✔ Solve radical equations in one variable..

✔ Explain how extraneous solutions may arise when solving radical equations.

✔ Solve radical inequalities and apply the solution within a real-world context.

Essential Understanding

Solving equations that include radicals or rational exponents is similar to solving rational equations, except that rather than eliminating the rational expression through multiplication, the radical expression is eliminated by raising both sides of the equation to the power equal to the index of the radical and applying the Power of a Power Property.

COHERENCE

Previously in this course, students:

- Solved rational equations and identified extraneous solutions.

In this lesson, students:

- Solve radical equations, identifying extraneous solutions.

Later in this topic, students will:

- Find and solve equations of inverse functions.

RIGOR

This lesson emphasizes a blend of *procedural skill and fluency* and *application*.

- Students solve radical equations by isolating the radical, eliminating the radical, solving the equation, and then eliminating extraneous solutions.
- Students apply what they know about radical equations to solve radical inequalities within the context of real-world problems, such as determining appropriate doses of medicine for ranges of body mass.

Glossary is available at SavvasRealize.com.

Vocabulary Builder

REVIEW VOCABULARY

English | *Spanish*

- **radical equation** | *ecuación radical*
- **rational number** | *rational equation*
- **rational exponent** | *exponent racional*
- **reciprocal power** | *poder recípro*

NEW VOCABULARY

- **extraneous solution** | *solución extrana*

VOCABULARY ACTIVITY

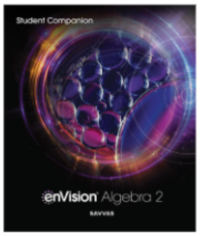
Help students understand the difference between radical equations and rational equations by helping them to understand the relationship between the root and suffix of each term. Both terms end in *-al*, meaning "pertaining to." In the case of *radical equation*, the root word is *radic*, meaning "root." In the case of *rational equation*, the root word is *ratio*, which is the quotient of two numbers.

Have students identify each equation as a radical equation or a rational equation.

$\sqrt[3]{x+6} = 42$ [radical equation] $\frac{1}{x+5} = 15$ [rational equation]

Student Companion

Students can do their work for the lesson in their *Student Companion* or on Savvas Realize.



Mathematics Overview

Students solve radical equations and equations with rational exponents by isolating the radical term, raising each side of the equation to a power to remove the radical, and identifying and eliminating any extraneous solutions. They then extend their understanding of solving radical equations to find the solution set of a radical inequality.

Applying Math Practices

Construct Viable Arguments

Students justify conclusions with mathematical ideas when they explain how a graph can be used to show the solution to a radical equation.

Look For and Make Use of Structure

Students identify that previously learned concepts and methods for solving equations are useful when solving inequalities.

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Activity

EXPLORE & REASON

INSTRUCTIONAL FOCUS Students explore solving an equation with exponents using two different methods. This prepares students to solve equations with radicals. Students learn that the methods used to solve equations with exponents and radicals are similar.

STUDENT COMPANION Students can complete the *Explore & Reason* activity in their *Student Companion*.

Before — WHOLE CLASS

Implement Tasks that Promote Reasoning and Problem Solving ETP

Q: What do you notice about the equations in Parts A and B? [Each equation contains the same integers and operations. However, in part A, you find $(a + 1)^2$, whereas in Part B, you find $\sqrt{(a + 1)}$.]

During — SMALL GROUP

Support Productive Struggle in Learning Mathematics ETP

Q: How can your knowledge of solving equations with exponents assist you in solving equations with radicals? [In order to isolate the expression with the variable, you need to perform the opposite operation. This means if the expression has a square root, you need to square it to isolate the variable.]

For Early Finishers

Q: Solve $5(x - 2)^4 - 7 = 73$ and $5\sqrt[4]{x - 2} - 7 = 73$. Which solution method did you choose? [$x = 0, 4$; $x = 65{,}538$; Each equation can be solved algebraically in just a few steps. It would be difficult to solve these equations by graphing because the numbers get very large.]

After — WHOLE CLASS

Facilitate Meaningful Mathematical Discourse ETP

Q: What methods can you use to solve an equation with a radical and an equation with an exponent? [For each type of equation, you can isolate the variables by performing the opposite operations. You can also graph the two equations created from the original equation.]

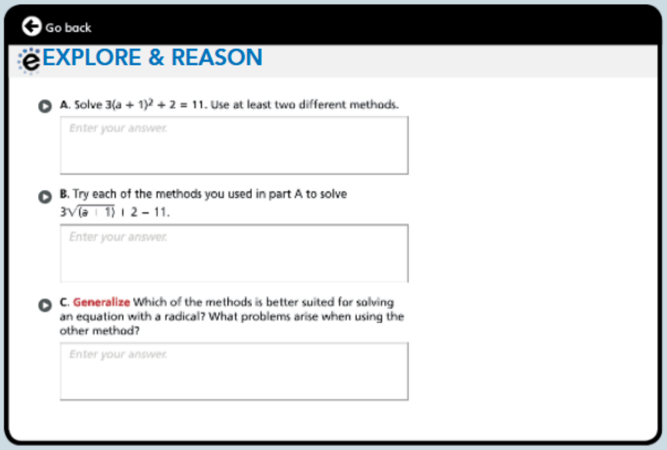
Use with **EXPLORE & REASON**

HABITS OF MIND

Construct Arguments The squares of two numbers are equal. Does that mean that the two numbers themselves must also be equal? Explain.

[No; two numbers that have the same value but opposite signs have the same square.]

Critique & Explain is available at SavvasRealize.com.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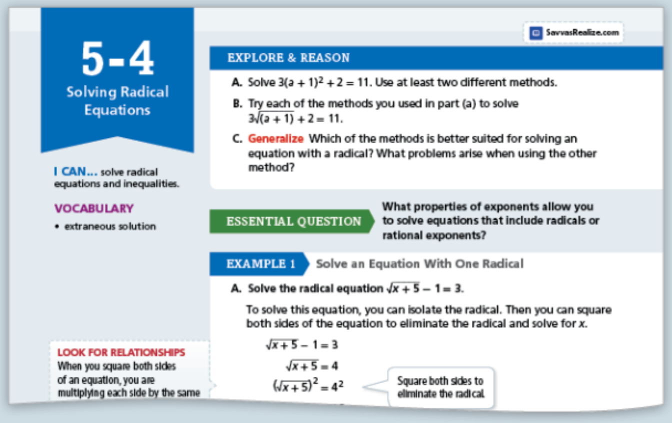
EXPLORE & REASON
A. Solve $3(a + 1)^2 + 2 = 11$. Use at least two different methods.
Enter your answer.
B. Try each of the methods you used in part A to solve $3\sqrt{(a + 1)} + 2 = 11$.
Enter your answer.
C. Generalize Which of the methods is better suited for solving an equation with a radical? What problems arise when using the other method?
Enter your answer.

STUDENT EDITION


5-4 Solving Radical Equations
I CAN… solve radical equations and inequalities.
VOCABULARY
• extraneous solution
EXPLORE & REASON
A. Solve $3(a + 1)^2 + 2 = 11$. Use at least two different methods.
B. Try each of the methods you used in part (a) to solve $3\sqrt{(a + 1)} + 2 = 11$.
C. Generalize Which of the methods is better suited for solving an equation with a radical? What problems arise when using the other method?
ESSENTIAL QUESTION What properties of exponents allow you to solve equations that include radicals or rational exponents?
EXAMPLE 1 Solve an Equation With One Radical
A. Solve the radical equation $\sqrt{x + 5} - 1 = 3$.
To solve this equation, you can isolate the radical. Then you can square both sides of the equation to eliminate the radical and solve for x.
$\sqrt{x + 5} - 1 = 3$
$\sqrt{x + 5} = 4$
$(\sqrt{x + 5})^2 = 4^2$
Square both sides to eliminate the radical.
LOOK FOR RELATIONSHIPS When you square both sides of an equation, you are multiplying each side by the same

SAMPLE STUDENT WORK

A. $a = -1 \pm \sqrt{3}$; You can solve this by isolating $(a + 1)^2$ and then taking the square root of each side. Or you could graph each side of the equation on the same coordinate grid and find the intersection points.

B. $a = 8$

C. Solving algebraically seems easier for an equation with a radical. Graphing can be difficult without technology, and it may only yield an approximate answer.

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Activity

INTRODUCE THE ESSENTIAL QUESTION

Establish Mathematics Goals to Focus Learning ETP

Introduce students to solving equations with radicals and rational exponents. Remind them that, just as they used properties of exponents to solve equations with exponents, they can use these properties to solve equations with radicals.

EXAMPLE 1 Solve an Equation With One Radical

Pose Purposeful Questions ETP

Q: What does it mean to *isolate* the *radical*? [Rewrite the equation so the radical is alone on one side of the equation.]

Q: How do you eliminate the square root or cube root? Why do you want to eliminate them? [First isolate the radical on one side of the equation. Then perform the opposite operation (square or cube) on both sides of the equation. Eliminating the radical allows you to solve for x.]

Try It! Answers

1. a. $x = 6$ **b.** $x = 9$

Elicit and Use Evidence of Student Thinking ETP

Q: How do you know whether to square or cube the radical when solving for x? [If the equation contains a square root, then square both sides to eliminate the radical. If the equation contains a cube root, then cube both sides to eliminate the radical.]

Examples are available at SavvasRealize.com.

equations and inequalities.

VOCABULARY
- extraneous solution

ESSENTIAL QUESTION What properties of exponents allow you to solve equations that include radicals or rational exponents?

EXAMPLE 1 Solve an Equation With One Radical

A. Solve the radical equation $\sqrt{x+5} - 1 = 3$.

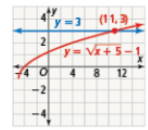
To solve this equation, you can isolate the radical. Then you can square both sides of the equation to eliminate the radical and solve for x.

$$\sqrt{x+5} - 1 = 3$$
$$\sqrt{x+5} = 4$$
$$(\sqrt{x+5})^2 = 4^2$$
Square both sides to eliminate the radical.
$$x + 5 = 16$$
$$x = 11.$$

LOOK FOR RELATIONSHIPS When you square both sides of an equation, you are multiplying each side by the same quantity. The expression for the quantity differs, but $\sqrt{x+5}$ and 4 are equal.

So the solution to the radical equation is $x = 11$.

You can check by graphing $y = \sqrt{x+5} - 1$ and $y = 3$ on the same coordinate axes. The graphs of the equations intersect at (11, 3).



B. Solve the radical equation $\sqrt[3]{x} + 2 = 4$.

$$\sqrt[3]{x} + 2 = 4$$
$$\sqrt[3]{x} = 2$$
$$(\sqrt[3]{x})^3 = 2^3$$
Cube both sides to eliminate the cube root.
$$x = 8$$

So the solution to the radical equation is $x = 8$.
Check your answer by substituting 8 for x in the original equation:
$\sqrt[3]{8} + 2 = 2 + 2 = 4$

Try It! 1. Solve each radical equation. Check your solution by substituting into the given equation or graphing.
a. $\sqrt{x-2} + 3 = 5$ b. $\sqrt[3]{x-1} = 2$

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Additional Examples

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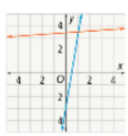
ADDITIONAL EXAMPLE 3 ESSENTIAL QUESTION

The following functions are shown in the coordinate plane.

$y = \sqrt{\frac{1}{2}x + 12}$ $y = 6x - 3$

Find the intersection of the graphs.



Solve $\sqrt{\frac{1}{2}x + 12} = 6x - 3$.

$x \approx 1.09$ and -0.08

-0.08 is an extraneous solution, so $x \approx 1.09$.

The intersection of the graphs is at approximately (1.09, 3.54).

SOLUTION

Example 3 Help students understand how to use what they know about radical equations to solve a system of consisting of a linear and a radical equation.

Q: How can you use the graph to predict whether there will be any extraneous solutions?
[The graphs of the functions intersect at 1 point, so I expect to find 1 solution when I solve the equation. If I find 2 solutions, it is likely that 1 will be extraneous.]

Additional Examples are available at SavvasRealize.com.

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ADDITIONAL EXAMPLE 6 ESSENTIAL QUESTION

An amusement park ride swings like a pendulum. A pendulum's swing is represented by the formula $S = 2\pi\sqrt{\frac{L}{32}}$. S is the time, in seconds, the pendulum takes to swing back and forth, and L is the length of the pendulum in feet.

The engineers at the amusement park have decided that the bottom of the ride needs to be 3 feet above the ground and that it should take a person no more than 5.15 seconds to swing back and forth. How high off the ground can the top of the ride be? Use 3.14 for π. Round your answer to the nearest hundredth.

Write an inequality to represent the time.

$$S \le 5.15$$
$$2\pi\sqrt{\frac{L}{32}} \le 5.15$$

Let H represent the height of the top of the ride, in feet.
Then $H = L + 3$, so $L = H - 3$.

$$2\pi\sqrt{\frac{H-3}{32}} \le 5.15$$
$$\sqrt{\frac{H-3}{32}} \le \frac{5.15}{2\pi}$$

TRY ANOTHER ONE

Example 6 Help students understand how to solve real-world inequalities.

Q: How do you know what to solve for when you are finding how high off the ground the ride should be? [The pendulum needs to be hung so that the bottom of it is 3 feet off the ground, so you need to solve for H, or $L + 3$.]

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Activity

EXAMPLE 2 Rewrite a Formula

Pose Purposeful Questions ETP

Q: Why is the equation solved for y?

[The equation is solved for y so you can find the height of the suspension cable more easily when you know your distance from the middle of the bridge.]

Q: Why are parentheses placed around $0.010583x$ in the third step?

[Parentheses are placed around $0.010583x$ to ensure that you square both 0.010583 and x.]

Try It! Answer

2. $d = \dfrac{v^2}{12.74}$

Common Error

Try It! 2 After dividing by the constant, some students may forget to square both the variable and the constant, which would result in $\frac{v^2}{3.57} = d$ instead of $\frac{v^2}{12.74} = d$. Have students make sure the entire equation has been squared, including 3.57.

HABITS OF MIND

Use with EXAMPLES 1 & 2

Reason Reese solved the equation in Try It 1a by first squaring both sides. Is this an appropriate first step? Why or why not?

[No; you must first subtract 3 from both sides of the equation to isolate the radical. Then you can square both sides.]

Examples are available at SavvasRealize.com.

APPLICATION

EXAMPLE 2 Rewrite a Formula

The suspension cables from the Golden Gate Bridge's towers are farther above the roadway near the towers and closer to the roadway near the middle of the bridge. You can figure out your distance from the middle of the bridge, x, in feet, and the height of the suspension cable, y, in feet, by using the given equation. About how far is the cable from the roadway when you are 200 ft from the middle of the bridge?

Rewrite the equation to isolate y so you can express the vertical distance between the roadway of the bridge and the suspension cable in terms of x.

$$x = \frac{\sqrt{y-220}}{0.010583}$$

$$0.010583x = \sqrt{y-220}$$

$$(0.010583x)^2 = \left(\sqrt{y-220}\right)^2$$

$$0.000112x^2 \approx y - 220$$

$$0.000112x^2 + 220 \approx y$$

This equation gives the vertical distance between the deck and the suspension cable for different values of x.

$$0.000112(200)^2 + 220 \approx y$$

$$224.48 \approx y$$

The cable is about 224 ft above the bridge's roadway when you are 200 ft from the middle of the bridge.

COMMON ERROR
When squaring a term that includes a coefficient and a variable, remember to square both parts of the term. If you have to round the coefficient after squaring, be sure to use the approximately equal sign.

Try It! 2. The speed, v, of a vehicle in relation to its stopping distance, d, is represented by the equation $v = 3.57\sqrt{d}$. What is the equation for the stopping distance in terms of the vehicle's speed?

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ELL English Language Learners (*Use with* EXAMPLE 2)

SPEAKING BEGINNING In small groups, have students isolate an object from a group of objects. Have students discuss what the term *isolate* means (to set apart or to be alone). If they have not heard the word before, challenge them to use it in a sentence based on the definition given.

Q: When have you heard the word, or a form of the word, *isolate* used?

[The prisoner was put in solitary isolation. The patient with an infectious disease must be isolated, even from his family.]

Q: How are your answers to the last question related to *isolating* the variable y in the example?

[Isolating y means to rewrite the equation so y is alone on one side.]

READING DEVELOPING Ask students to read the first sentence of the example and read the definition of suspend from the dictionary.

Q: Can you identify the root word in the word *suspension*? [*suspend*]

Q: What does *suspend* mean? [to hang by attachment to something above]

Q: Have the definitions of *suspension* and *suspension bridge* on index cards. Hand out the cards and have students read the definitions aloud. How do the definitions help you to understand the meaning of suspension cables mentioned in the example? [A *suspension bridge* is a bridge suspended by cables anchored on the ends and raised by towers, so the *suspension cables* are the cords from which the Golden Gate Bridge hangs.]

LISTENING EXPANDING Have students listen as you read the following sentences aloud. 1) He wore a pair of glasses on the *bridge* of his nose. 2) The isthmus is a land *bridge* between the two countries. 3) The interpreter is a *bridge* between people who speak different languages.

Q: What does the word *bridge* indicate in each of these examples?

[a connection between two things]

Q: The example describes the physical structure of a bridge. Does a bridge always have to be a structure that you physically cross over? Explain.

[No; the interpreter is acting as a different kind of bridge.]

EXAMPLE 3 Identify an Extraneous Solution

Build Procedural Fluency From Conceptual Understanding ETP

PART A

Q: Why can you find potential solutions to an equation that are not actual solutions of the original equation?
[Squaring both sides of an equation may create a new equation with two solutions. Only one of these solutions may satisfy the original equation.]

PART B

Q: When solving an equation, what indicates that there may be an extraneous solution?
[When the solution steps require you to solve a quadratic equation, it indicates there may be an extraneous solution.]

Try It! Answers

3. a. $x = 8$; $x = -1$ is an extraneous solution
   b. $x = -1$, $x = -2$

Elicit and Use Evidence of Student Thinking ETP

Q: In part (b), why do you need to be careful when squaring both sides of the equation?
[You need to make sure that you square the entire left side, $(x + 2)$. This will require applying the Distributive Property to multiply the two binomials.]

Examples are available at SavvasRealize.com.

CONCEPTUAL UNDERSTANDING

EXAMPLE 3 Identify an Extraneous Solution

A. Solve the radical equation $\sqrt{3x-2} = x - 4$.

$$\sqrt{3x-2} = x - 4$$
$$\left(\sqrt{3x-2}\right)^2 = (x-4)^2$$
$$3x - 2 = x^2 - 8x + 16$$
$$0 = x^2 - 11x + 18$$
$$0 = (x-9)(x-2)$$
$$x = 9 \text{ or } x = 2$$

Check the potential solutions by substituting the solutions for x in the original equation:

$$\sqrt{3x-2} = x - 4 \qquad \sqrt{3x-2} = x - 4$$
$$\sqrt{3(9)-2} = 9 - 4 \qquad \sqrt{3(2)-2} = 2 - 4$$
$$\sqrt{25} = 5 \checkmark \qquad \sqrt{4} \neq -2 \times$$

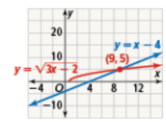
So 9 is the only solution to the equation and 2 is an extraneous solution.

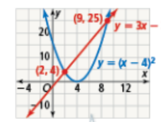
VOCABULARY
An **extraneous solution** is a solution of an equation *derived* from an equation, but it is *not* a solution of the original equation.

B. Why does this extraneous solution arise?

The first step in solving the equation in part (a) was to square both sides.

Graph on one grid, $y = \sqrt{3x-2}$ and $y = x - 4$. Then graph on another grid $y = 3x - 2$ and $y = (x-4)^2$.





These intersect at (9, 5), so the solution to $\sqrt{3x-2} = x - 4$ is $x = 9$.

These intersect at (2, 4) and (9, 25), so the solutions to $\left(\sqrt{3x-2}\right)^2 = (x-4)^2$ are $x = 2$ and $x = 9$.

This means that the equations $\sqrt{3x-2} = x - 4$ and $\left(\sqrt{3x-2}\right)^2 = (x-4)^2$ are not equivalent equations. Both graphs have an intersection point at $x = 9$, but the second graph also has an intersection point at $x = 2$. Squaring both sides of the original equation created an extraneous solution.

Try It! 3. Solve each radical equation. Check for any extraneous solutions.

a. $x = \sqrt{7x+8}$ b. $x + 2 = \sqrt{x+2}$

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Activity

EXAMPLE 4 Solve Equations With Rational Exponents

Pose Purposeful Questions ETP

PART A

Q: Why do you raise each side of an equation to the reciprocal power?
[Multiplying a rational exponent by its reciprocal equals 1.]

PART B

Q: Why did the right side of the equation change from $(x-2)^3$ to $(x-2)^2$?
[Each side of the equation was raised to the reciprocal power of the left side, so $\frac{3}{1} \cdot \frac{2}{3} = 2$.]

Try It! Answers

4. a. $x = 5, -2$
   b. $x = 9$

Elicit and Use Evidence of Student Thinking ETP

Q: Why do you isolate the expression with the rational exponent?
[When you isolate the expression with the rational exponent, only the expression will remain after each side of the equation is raised to the reciprocal power.]

Examples are available at SavvasRealize.com.

EXAMPLE 4 Solve Equations With Rational Exponents

A. What are the solutions to the equation $(x^2+5x+5)^{\frac{3}{2}} = 1$?

$$(x^2+5x+5)^{\frac{3}{2}} = 1$$
$$\left((x^2+5x+5)^{\frac{3}{2}}\right)^{\frac{2}{3}} = (1)^{\frac{2}{3}}$$ Raise both sides to the reciprocal power.
$$x^2+5x+5 = 1$$
$$x^2+5x+4 = 0$$
$$(x+4)(x+1) = 0$$
$$x+4=0 \text{ or } x+1=0$$ Use the Zero-Product Property.
$$x=-4 \text{ or } x=-1$$

STUDY TIP
An equation can have one solution, multiple solutions, or no solutions, so it's important to check all potential solutions.

Check for extraneous solutions.

$$\left((-4)^2+5(-4)+5\right)^{\frac{3}{2}} \stackrel{?}{=} 1 \qquad \left((-1)^2+5(-1)+5\right)^{\frac{3}{2}} \stackrel{?}{=} 1$$
$$(16-20+5)^{\frac{3}{2}} \stackrel{?}{=} 1 \qquad (1-5+5)^{\frac{3}{2}} \stackrel{?}{=} 1$$
$$(1)^{\frac{3}{2}} \stackrel{?}{=} 1 \qquad (1)^{\frac{3}{2}} \stackrel{?}{=} 1$$
$$1 = 1 \checkmark \qquad 1 = 1 \checkmark$$

This equation has two solutions, $x=-4$ and $x=-1$. There are no extraneous solutions.

B. What is the solution to $(x+18)^{\frac{3}{2}} = (x-2)^3$?

$$(x+18)^{\frac{3}{2}} = (x-2)^3$$
$$\left((x+18)^{\frac{3}{2}}\right)^{\frac{2}{3}} = \left((x-2)^3\right)^{\frac{2}{3}}$$ Raise both sides to the reciprocal power.
$$x+18 = (x-2)^2$$
$$x+18 = x^2-4x+4$$
$$x^2-5x-14 = 0$$
$$(x+2)(x-7) = 0$$
$$x+2=0 \qquad x-7=0$$ Use the Zero-Product Property.
$$x=-2 \qquad x=7$$

Check for extraneous solutions.

$$(-2+18)^{\frac{3}{2}} \stackrel{?}{=} (-2-2)^3 \qquad (7+18)^{\frac{3}{2}} \stackrel{?}{=} (7-2)^3$$
$$16^{\frac{3}{2}} \stackrel{?}{=} (-4)^3 \qquad 25^{\frac{3}{2}} \stackrel{?}{=} 5^3$$
$$64 \neq -64 ✗ \qquad 125 = 125 \checkmark$$

So 7 is the only solution to the equation, and -2 is an extraneous solution.

Try It! 4. Solve each equation.
a. $(x^2-3x-6)^{\frac{3}{2}} - 14 = -6$ b. $(x-8)^{\frac{3}{2}} = (10-x)^3$

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Extend Student Thinking

USE WITH EXAMPLE 4 Have students explore solving equations with rational exponents that have no solution by graphing. Students should recognize that $\sqrt{\square}$ is equal to an expression raised to the $\frac{1}{2}$ power.

Solve the equation $\sqrt{6x+8} = (x-7)^{\frac{1}{2}}$.

Q: What strategy can you use to solve this equation?
[Sample: graphing]

Q: What do you recognize about the solution?
[There is no real solution. The graphs of $y = \sqrt{6x+8}$ and $y = (x-7)^{\frac{1}{2}}$ do not intersect. Also, when you solve for x, you find $x = -3$, and because -3 is not in the domain of either expression, it is an extraneous solution.]

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Activity

EXAMPLE 5 Solve an Equation With Two Radicals

Use and Connect Mathematical Representations ETP

Q: What is another strategy you could use to find the solution to an equation without doing calculations?
[You could graph the two sides of the equation separately. The solution is where the graphs intersect.]

Q: Why does a radical remain in the solution steps after the two sides of the equation have been squared the first time?
[After expanding the right side using the Distributive Property, an integer multiplied by a radical will contain a radical.]

Try It! Answers

5. **a.** $x = 12$; $x = 0$ is an extraneous solution

   b. $x = 6$; $x = \frac{6}{49}$ is an extraneous solution

Elicit and Use Evidence of Student Thinking ETP

Q: What method can you use to solve equations with two radicals?
[Isolate the radicals on either side of the equation and square both sides. Then the first equation can be solved by factoring, but the second equation must be solved using the Quadratic Formula.]

HABITS OF MIND

Use with EXAMPLES 3, 4, & 5

Reason Why are extraneous solutions a possibility for radical equations?

[Extraneous solutions can occur when you square both sides of the equation.]

Examples are available at SavvasRealize.com.

EXAMPLE 5 Solve an Equation With Two Radicals

Solve the radical equation $\sqrt{x+9} - \sqrt{2x} = 3$.

When an equation has two radicals, start the solution process by isolating one radical.

COMMON ERROR
A common mistake when squaring an expression like $\sqrt{2x} + 3$ is to only square the radical portion. Square this expression as you would any binomial, by using the Distributive Property.

$$\sqrt{x+9} - \sqrt{2x} = 3$$

$$\sqrt{x+9} = \sqrt{2x} + 3 \quad \text{Isolate one radical sign.}$$

$$(\sqrt{x+9})^2 = (\sqrt{2x} + 3)^2$$

$$x + 9 = 2x + 6\sqrt{2x} + 9$$

$$0 = x + 6\sqrt{2x}$$

$$-x = 6\sqrt{2x} \quad \text{Isolate the remaining radical and square again.}$$

$$(-x)^2 = (6\sqrt{2x})^2$$

$$x^2 - 72x = 0$$

$$x(x - 72) = 0$$

$$x = 0 \text{ or } x = 72$$

Check the potential solutions by substituting 0 and 72 for x in the original equation:

$$\sqrt{x+9} - \sqrt{2x} \stackrel{?}{=} 3 \qquad \sqrt{x+9} - \sqrt{2x} \stackrel{?}{=} 3$$

$$\sqrt{(0)+9} - \sqrt{2(0)} \stackrel{?}{=} 3 \qquad \sqrt{(72)+9} - \sqrt{2(72)} \stackrel{?}{=} 3$$

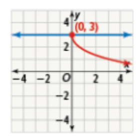
$$\sqrt{9} - \sqrt{0} \stackrel{?}{=} 3 \qquad \sqrt{81} - \sqrt{144} \stackrel{?}{=} 3$$

$$3 - 0 \stackrel{?}{=} 3 \qquad 9 - 12 \stackrel{?}{=} 3$$

$$3 = 3 \checkmark \qquad -3 \neq 3 \times$$

The only solution is 0. The value 72 does not make the original equation true, so it is an extraneous solution.

You can also see this by graphing $y = \sqrt{x+9} - \sqrt{2x}$ and $y = 3$ together. They intersect only at $x = 0$.



Try It! 5. Solve each radical equation. Check for extraneous solutions.

a. $\sqrt{x+4} - \sqrt{3x} = -2$ b. $\sqrt{15-x} - \sqrt{6x} = -3$

Support Struggling Students

USE WITH EXAMPLE 5 ome students may struggle with squaring each side of an equation when solving an equation with two radicals. Have students practice squaring expressions.

- Square the following expressions.

Q: 12
[144]

Q: $\sqrt{5x}$
[$5x$]

Q: $\sqrt{x} + 6$
[$x + 12\sqrt{x} + 36$]

Q: $\sqrt{3x} + 10$
[$3x + 20\sqrt{3x} + 100$]

Q: What is important to remember when squaring these expressions?
[Square the entire expression, not just what is under the radical.]

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Activity

EXAMPLE 6 Solve a Radical Inequality

Pose Purposeful Questions ETP

Q: Why is the inequality written as < 1.9?

[This dosage is for a patient whose BSA is less than 1.9.]

Q: How do the steps to solve the problem change if you are given the height instead of the mass?

[The steps are similar; the only difference is that you substitute for M and solve for H in this case.]

Try It! Answer

6. less than 155.52 cm tall

Elicit and Use Evidence of Student Thinking ETP

Q: How can this inequality be written according to these new conditions?

[$\sqrt{\frac{H \cdot 75}{3{,}600}} < 1.8$]

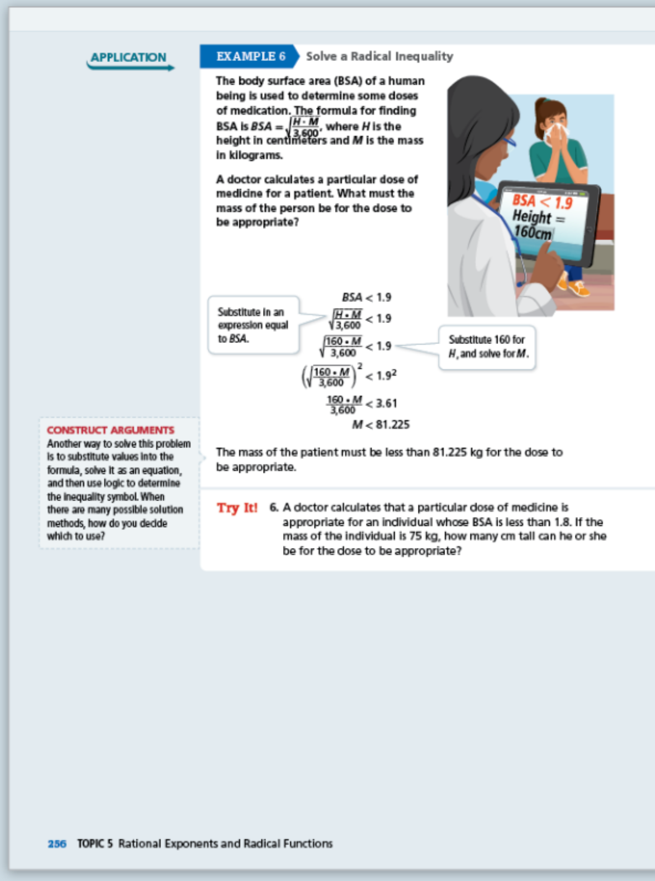
HABITS OF MIND

Use with **EXAMPLE 6**

Make Sense and Persevere How are the steps for solving a radical inequality different from the steps for solving a radical equation?

[The steps for solving a radical inequality are the same as the steps for solving a radical equation. The only time they would be different is when multiplying or dividing each side of the inequality by a negative number; then you have to remember to reverse the inequality symbol.]

Examples are available at SavvasRealize.com.



APPLICATION

EXAMPLE 6 Solve a Radical Inequality

The body surface area (BSA) of a human being is used to determine some doses of medication. The formula for finding BSA is $BSA = \sqrt{\frac{H \cdot M}{3{,}600}}$, where H is the height in centimeters and M is the mass in kilograms.

A doctor calculates a particular dose of medicine for a patient. What must the mass of the person be for the dose to be appropriate?

$$BSA < 1.9$$

$$\sqrt{\frac{H \cdot M}{3{,}600}} < 1.9$$

Substitute in an expression equal to BSA.

$$\sqrt{\frac{160 \cdot M}{3{,}600}} < 1.9$$

Substitute 160 for H, and solve for M.

$$\left(\sqrt{\frac{160 \cdot M}{3{,}600}}\right)^2 < 1.9^2$$

$$\frac{160 \cdot M}{3{,}600} < 3.61$$

$$M < 81.225$$

The mass of the patient must be less than 81.225 kg for the dose to be appropriate.

CONSTRUCT ARGUMENTS Another way to solve this problem is to substitute values into the formula, solve it as an equation, and then use logic to determine the inequality symbol. When there are many possible solution methods, how do you decide which to use?

Try It! 6. A doctor calculates that a particular dose of medicine is appropriate for an individual whose BSA is less than 1.8. If the mass of the individual is 75 kg, how many cm tall can he or she be for the dose to be appropriate?

256 TOPIC 5 Rational Exponents and Radical Functions

CONCEPT SUMMARY Solving Radical Equations

Q: What are two methods you can use to check for extraneous solutions?

[You can substitute your solutions in the original equation, or you can graph the expression on each side of the equation and find the intersection.]

Do You UNDERSTAND? | Do You KNOW HOW?

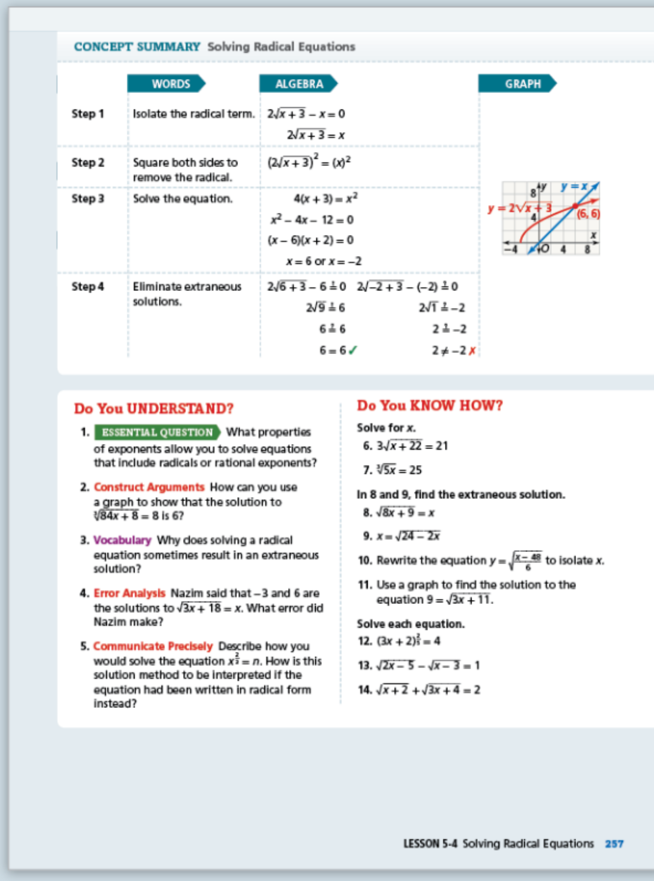
Common Error

Exercise 12 Students may forget to multiply by the reciprocal power and may multiply both sides by the exponent. In this case, they may think $3x + 2 = 4^{\frac{2}{5}}$. Encourage students to make sure, when they multiply by an exponent, that the exponent will equal 1, $\frac{2}{5} \cdot \frac{5}{2} = 1$ and $\frac{2}{5} \cdot \frac{2}{5} = \frac{4}{5}$.

Answers

1. You can use Power of a Power Property to eliminate radicals and rational exponents.
2. Graph $y = \sqrt[3]{84x + 8}$ and $y = 8$. The graphs will intersect at $x = 6$.
3. Squaring both sides of an equation when solving sometimes produces an extraneous solution.
4. Nazim didn't check for extraneous solutions. The real solution is 6; -3 is extraneous.
5. Raise both sides to the power $\frac{3}{2}$; cube both sides and then find the square root of the result.
6. $x = 27$
7. $x = 3{,}125$
8. $x = -1$
9. $x = -6$
10. $x = 6y^2 + 48$
11. $x = 23.\overline{3}$
12. $x = 10$
13. $x = 7$ or $x = 3$
14. $x = -1$

Concept Summary, Do You Understand, and **Do You Know How** are available at SavvasRealize.com.



CONCEPT SUMMARY Solving Radical Equations

	WORDS	ALGEBRA
Step 1	Isolate the radical term.	$2\sqrt{x+3} - x = 0$ $2\sqrt{x+3} = x$
Step 2	Square both sides to remove the radical.	$(2\sqrt{x+3})^2 = (x)^2$
Step 3	Solve the equation.	$4(x+3) = x^2$ $x^2 - 4x - 12 = 0$ $(x-6)(x+2) = 0$ $x = 6$ or $x = -2$
Step 4	Eliminate extraneous solutions.	$2\sqrt{6+3} - 6 \stackrel{?}{=} 0$ $\quad 2\sqrt{-2+3} - (-2) \stackrel{?}{=} 0$ $2\sqrt{9} \stackrel{?}{=} 6$ $\quad 2\sqrt{1} \stackrel{?}{=} -2$ $6 \stackrel{?}{=} 6$ $\quad 2 \stackrel{?}{=} -2$ $6 = 6$ ✓ $\quad 2 \neq -2$ ✗

GRAPH

Do You UNDERSTAND?

1. ESSENTIAL QUESTION What properties of exponents allow you to solve equations that include radicals or rational exponents?
2. Construct Arguments How can you use a graph to show that the solution to $\sqrt[3]{84x + 8} = 8$ is 6?
3. Vocabulary Why does solving a radical equation sometimes result in an extraneous solution?
4. Error Analysis Nazim said that -3 and 6 are the solutions to $\sqrt{3x + 18} = x$. What error did Nazim make?
5. Communicate Precisely Describe how you would solve the equation $x^{\frac{2}{3}} = n$. How is this solution method to be interpreted if the equation had been written in radical form instead?

Do You KNOW HOW?

Solve for x.

6. $3\sqrt{x + 22} = 21$
7. $\sqrt[3]{5x} = 25$

In 8 and 9, find the extraneous solution.

8. $\sqrt{8x + 9} = x$
9. $x = \sqrt{24 - 2x}$
10. Rewrite the equation $y = \sqrt{\frac{x - 48}{6}}$ to isolate x.
11. Use a graph to find the solution to the equation $9 = \sqrt{3x + 11}$.

Solve each equation.

12. $(3x + 2)^{\frac{2}{5}} = 4$
13. $\sqrt{2x - 5} - \sqrt{x - 3} = 1$
14. $\sqrt{x + 2} + \sqrt{3x + 4} = 2$

LESSON 5-4 Solving Radical Equations 257

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Practice

PRACTICE & PROBLEM SOLVING

Practice & Problem Solving and **video tutorials** are available at SavvasRealize.com.

Lesson Practice You may opt to have students complete the automatically scored Practice and Problem Solving items online powered by MathXL for School.

Choose from: **Lesson Practice**
Adaptive Practice

You may also take advantage of the bank of exercises for assigning additional practice.

Assignment Guide

On-Level	Advanced
15, 17, 18, 21–46	15–46

Engage Through Student Choice

Promote student agency by allowing students to choose practice items. You may structure this choice in many ways.
For example:
Assign each section a point value. Students choose at least one item from each section and items chosen should have a minimum of 20 total points.

Understand, Apply....................2 points each
Practice.....................................1 point each
Assessment Practice.................1 point each
Performance Task ....3 points

Item Analysis

Example	Items	DOK
1	21–24, 44	1
	18	2
2	25–28	1
	16, 17, 40, 43, 46	2
3	29–32	1
	15	2
4	33–35, 45	1
	19, 41	3
5	36–38	1
	20	2
6	39	1
	42	2

Scan the QR code to access a video tutorial.

PRACTICE & PROBLEM SOLVING

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UNDERSTAND

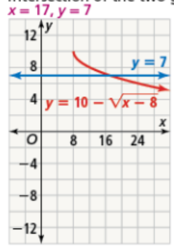
15. **Generalize** Explain how to identify an extraneous solution for an equation containing a radical expression.

16. **Look for Relationships** Write a radical equation that relates a square's perimeter to its area. Explain your reasoning. Use s to represent the side length of the square.

17. **Error Analysis** Describe and correct the error a student made in rewriting the equation to isolate y.

$$x = \frac{\sqrt{58+y}}{1.98}$$
$$1.98x = \sqrt{58+y}$$
$$1.98x^2 = 58 + y$$
$$1.98x^2 - 58 = y \quad ✗$$

18. **Use Appropriate Tools** Find the point of intersection of the two graphs. $x = 17, y = 7$



19. **Higher Order Thinking** Some, but not all, equations with rational exponents have extraneous solutions. What is the relationship between the exponents and the possibility of having extraneous solutions for equations with rational exponents? Explain your reasoning.

20. **Communicate Precisely** Describe the process used to solve an equation with two radical expressions. How is this process different from solving an equation with only one radical expression?

PRACTICE

Solve each radical equation. SEE EXAMPLE 1

21. $\sqrt[3]{x} + 8 = 13$ 125
22. $\sqrt{4x} = 11$ 30.25
23. $\sqrt{75 + x} - 6 = 14$ 325
24. $25 - \sqrt[4]{x} = 22$ 81

Solve for y. SEE EXAMPLE 2

25. $x = 3(\sqrt[3]{15 + y})$
26. $x = \frac{\sqrt{2y}}{26}$ $y = 338x^2$
27. $x = \frac{\sqrt{y - 14.2}}{0.05}$ $y = 0.0025x^2 + 14.2$
28. $x = \frac{1}{3}(\sqrt[4]{y})$ $y = 81x^4$

Solve each radical equation. Check for extraneous solutions. SEE EXAMPLE 3

29. $x = \sqrt{x + 6}$ $x = 3$
30. $2x = \sqrt{17x - 15}$ $x = \frac{5}{4}$ and $x = 3$
31. $4x = \sqrt{6x + 10}$ $x = 1$
32. $x = \sqrt{56 - x}$ $x = 7$

Solve. SEE EXAMPLE 4

33. $0.5(x^2 + 5x + 136)^{\frac{2}{3}} = 50$ $x = 27$ and $x = -32$
34. $2(x^2 - 12x - 4)^{\frac{1}{2}} - 3 = 15$ $x = -5$ and $x = 17$
35. $(x^2 + 4x + 5)^{\frac{3}{2}} + 1 = 0$ There are no solutions.

Solve each radical equation. Check for extraneous solutions. SEE EXAMPLE 5

36. $\sqrt{6 + x} - \sqrt{x - 5} = 2$ $x = 8\frac{1}{16}$
37. $\sqrt{4x + 5} - \sqrt{x + 1} = 1$ $x = -1$ and $x = -\frac{5}{9}$
38. $\sqrt{x + 1} + 1 = \sqrt{x + 3}$ $x = -\frac{3}{4}$

Solve using the formula $BSA = \sqrt{\frac{H \cdot M}{3{,}600}}$. SEE EXAMPLE 6

39. A sports medicine specialist determines that a hot-weather training strategy is appropriate for this athlete. To the nearest tenth, what can the mass of the athlete be for the training strategy to be appropriate? 87.3 kg

258 TOPIC 5 Rational Exponents and Radical Functions

Answers

15. Substitute each potential solution into the original equation. If it makes the equation true, it is a real solution.

16. $P = 4\sqrt{A}$; Sample: If $A = s^2$, then $s = \sqrt{A}$. Substituting $\sqrt{A}$ for s in the perimeter formula results in the given answer.

17. The student squared x only instead of 1.98 and x. Here is the correct work:

$$1.98x = \sqrt{58 + y}$$
$$3.92x^2 = 58 + y$$
$$3.92x^2 - 58 = y$$

See answers for 19–20, 25 on the next page.

Answers

19. Only if the denominator of the rational exponent is even can there be extraneous solutions. Taking an even root of a negative number gives an extraneous solution, so if the root isn't even, there can't be an extraneous solution.

20. The process of solving an equation with two radicals is like solving an equation with one radical. Simplify the equation and square both sides to eliminate one radical; because there are two radicals, you have to square both sides a second time; Rewrite the equation in standard form, and then factor the equation using the Zero-Product Property to solve for the variable.

There are additional steps needed to eliminate the second radical.

25. $y = \frac{x^3}{27} - 15$

40. $d = \frac{s^2}{252.81f}$

41. about 5.44 mL

42. 13 ft

43. $\approx$ 3,000 ft

46. **Part A** $M = \frac{v^2 r}{2G}$

Part B $M \approx 5.99 \times 10^{24}$

Practice & Problem Solving is available at SavvasRealize.com.

PRACTICE & PROBLEM SOLVING

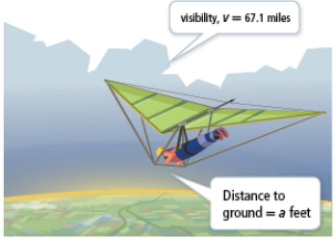
APPLY

40. Specialists can determine the speed a vehicle was traveling from the length of its skid marks, d, and the coefficient of friction, f. Rewrite the formula to solve for the length of the skid marks.

41. Make Sense and Persevere The half-life of a certain type of soft drink is 5 h. If you drink 50 mL of this drink, the formula $y = 50(0.5)^{\frac{t}{5}}$ tells the amount of the drink left in your system after t hours. How much of the soft drink will be left in your system after 16 hours?

42. Model With Mathematics Big Ben's pendulum takes 4 s to swing back and forth. The formula $t = 2\pi\sqrt{\frac{L}{32}}$ gives the swing time, t, in seconds, based on the length of the pendulum, L, in feet. What is the minimum length necessary to build a clock with a pendulum that takes longer than Big Ben's pendulum to swing back and forth?

43. Make Sense and Persevere Derek is hang gliding on a clear day. Use the formula for visibility, $v = 1.225\sqrt{a}$ to find the altitude at which Derek is hang gliding.



ASSESSMENT PRACTICE

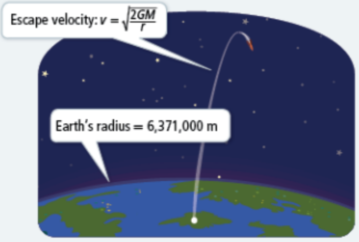
44. Complete the table to solve for the unknown value in the equation $y = \sqrt[3]{2x + z} - 12$, using the given values in each row.

y	x	z
0	462	804
−3	145	439
−10	1.25	5.5
3	1,679.5	16

45. SAT/ACT What is the solution to the equation $(x^2 + 5x + 25)^{\frac{3}{2}} = 343$?

Ⓐ −8 only
Ⓑ 3 only
Ⓒ 77 only
Ⓓ −8 and 3
Ⓔ There are no solutions.

46. Performance Task Escape velocity is the velocity at which an object must be traveling to leave a star or planet without falling back to its surface or into orbit. Escape velocity depends on the gravitational constant, $G = 6.67 \times 10^{-11}$, the mass, M in kilograms, and radius, r, of the star or planet.



Part A Rewrite the equation to solve for mass.

Part B Earth has an escape velocity of about 11,200 meters per second. What is Earth's mass in kilograms?

LESSON QUIZ

Use the Lesson Quiz to assess students' understanding of the mathematics in the lesson.

Students can take the Lesson Quiz online or you can download a printable copy from **SavvasRealize.com.** The Lesson Quiz is also available in the *Assessment Sourcebook*.

Item Analysis

Item	DOK	Skills Review and Practice
1	1	A35
2	2	A35
3	2	A35
4	2	A35
5	2	A35

RtI Use the student scores on the Lesson Quiz to prescribe differentiated assignments.

If students take the Lesson Quiz online, it will be automatically scored and appropriate differentiated practice will be assigned based on student performance.

I Intervention	0–3 points	• Reteach to Build Understanding • Mathematical Literacy and Vocabulary • Lesson Virtual Nerd videos
O On-Level	4 points	• Enrichment
A Advanced	5 points	• Enrichment

Lesson Quiz is available at SavvasRealize.com.

ASSESSMENT RESOURCES

Name ______

enVision Algebra 2
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5-4 Lesson Quiz

Solving Radical Equations

1. Solve the radical equation $\sqrt{x-1}+5=9$.

   $x =$ 17

2. **Part A**

   Leo solved the equation $b=\frac{1}{2}\sqrt[3]{a-1}$ for a, but made an error. His work is shown. Complete the sentences that follow.

   (i) $b=\frac{1}{2}\sqrt[3]{a-1}$

   (ii) $2b=\sqrt[3]{a-1}$

   (iii) $4b^2=a-1$

   (iv) $4b^2+1=a$

   The error is in line ☐ (i) ☐ (ii) ■ (iii) ☐ (iv).

   Part B

   What is the correct solution to the equation?

   Ⓐ $a=8b^3+1$ Ⓒ $a=8b^3$

   Ⓑ $a=8b^3-1$ Ⓓ $a=b^3+1$

3. Solve the equation $x+3=\sqrt{x+33}$.

   Ⓐ The equation has two solutions, −8 and 3.

   Ⓑ The solution to the equation is 3, and −8 is an extraneous solution.

   Ⓒ The solution to the equation is −8, and 3 is an extraneous solution.

   Ⓓ The equation has no real solutions.

4. What are the solutions to the equation $2(x^2+x-11)^{\frac{3}{2}}=54$?

   $x =$ 4 and $x =$ −5

5. Solve the radical equation $\sqrt{2x}-\sqrt{36-2x}=6$. Check for extraneous solutions.

   The solution is 18. The extraneous solution is 0.

enVision® Algebra 2 • Assessment Sourcebook

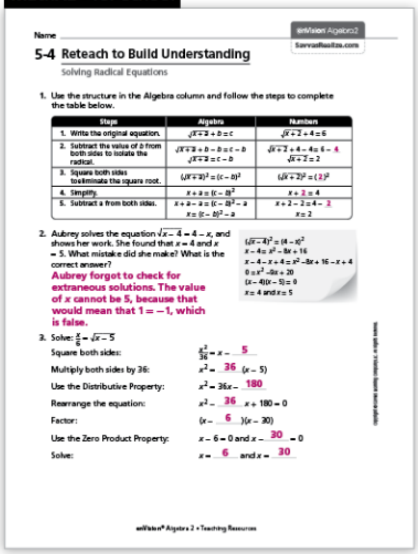
DIFFERENTIATED RESOURCES

I = Intervention   O = On-Level   A = Advanced

Reteach to Build Understanding I

Provides scaffolded reteaching for the key lesson concepts.

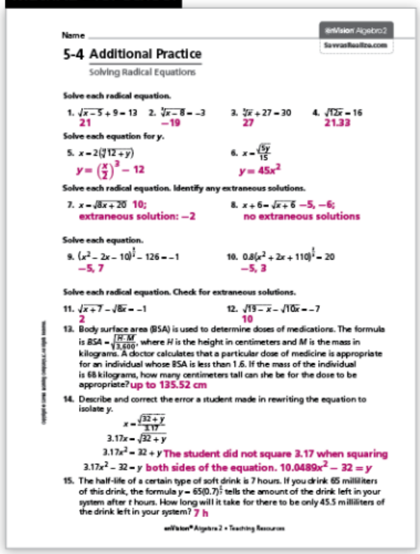
MathXL® for School


5-4 Reteach to Build Understanding
Solving Radical Equations

Additional Practice I O

Provides extra practice for each lesson.

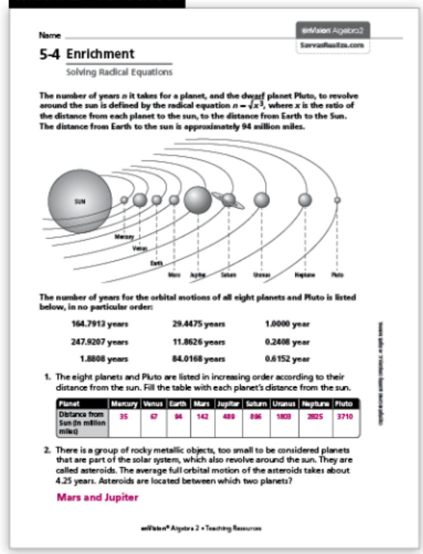
MathXL® for School


5-4 Additional Practice
Solving Radical Equations

Enrichment O A

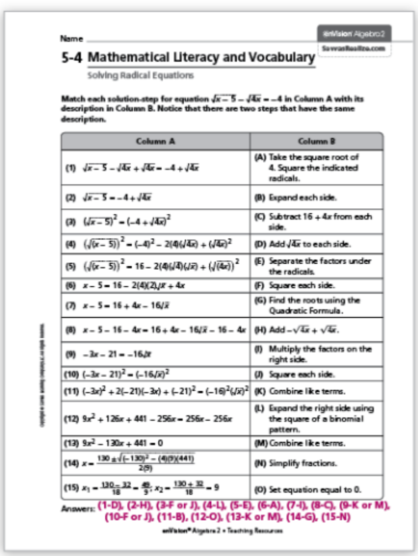
Presents engaging problems and activities that extend the lesson concepts.

MathXL® for School


5-4 Enrichment
Solving Radical Equations

Mathematical Literacy and Vocabulary I O

Helps students develop and reinforce understanding of key terms and concepts.


5-4 Mathematical Literacy and Vocabulary
Solving Radical Equations

Digital Differentiated Resources

The **Reteach to Build Understanding, Additional Practice,** and **Enrichment** activities are available as digital assignments. These activities are automatically assigned when students complete the lesson quiz online and are automatically scored.